## EVALUATION

### I. Multiple choice questions:

**1.** In hot summer after a bath, the body's
   a) internal energy decreases
   b) internal energy increases
   c) heat decreases
   d) no change in internal energy and heat

**2.** The graph between volume and temperature in Charles' law is
   a) an ellipse  b) a circle
   c) a straight line  d) a parabola

**3.** When a cycle tyre suddenly bursts, the air inside the tyre expands. This process is
   a) isothermal  b) adiabatic
   c) isobaric  d) isochoric

**4.** An ideal gas passes from one equilibrium state ($P_1$, $V_1$, $T_1$, $N$) to another equilibrium state ($2P_1$, $3V_1$, $T_2$, $N$). Then
   a) $T_1 = T_2$  b) $T_1 = \frac{T_2}{6}$
   c) $T_1 = 6T_2$  d) $T_1 = 3T_2$

**5.** When a uniform rod is heated, which of the following quantity of the rod will increase
   a) mass
   b) weight
   c) center of mass
   d) moment of inertia

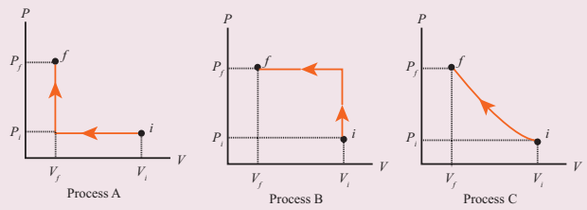
**6.** When food is cooked in a vessel by keeping the lid closed, after some time the steam pushes the lid outward. By considering the steam as a thermodynamic system, then in the cooking process
   a) $Q > 0$, $W > 0$,
   b) $Q < 0$, $W > 0$,
   c) $Q > 0$, $W < 0$,
   d) $Q < 0$, $W < 0$,

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**7.** When you exercise in the morning, by considering your body as thermodynamic system, which of the following is true?
   a) $\Delta U > 0$, $W > 0$,
   b) $\Delta U < 0$, $W > 0$,
   c) $\Delta U < 0$, $W < 0$,
   d) $\Delta U = 0$, $W > 0$,

**8.** A hot cup of coffee is kept on the table. After some time it attains a thermal equilibrium with the surroundings. By considering the air molecules in the room as a thermodynamic system, which of the following is true
   a) $\Delta U > 0$, $Q = 0$
   b) $\Delta U > 0$, $W < 0$
   c) $\Delta U > 0$, $Q > 0$
   d) $\Delta U = 0$, $Q > 0$

**9.** An ideal gas is taken from ($P_i$,$V_i$) to ($P_f$,$V_f$) in three different ways. Identify the process in which the work done on the gas the most.






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